SML Sampling

User manual for the script
SML_sampling_example_with_6_Italy_soil_units.R

Optimal selection of soil monitoring sites (Soil Monitoring Level) on a 1 km grid: customisation and execution guide

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1. Purpose and overview

The script selects the sampling sites of a soil monitoring network for a test area made of **6 domains** (soil unit x land use) grouped in **2 zones**: *Toscana meridionale* (60.4_Cropland, 60.4_Forest, 60.7_Cropland) and *Emilia orientale - delta del Po* (BE_Cropland, BA_Cropland, DO_Cropland).

The sample is optimised with the **SamplingStrata** package (genetic algorithm + Bethel allocation) so that the two target variables, soil pH (**PH**) and soil organic carbon (**SOC**), are estimated with a coefficient of variation (CV) of at most 5 % in every domain. Strata are built on two auxiliary variables: clay content (**ARG**) and elevation (**DEM**). Existing LUCAS points can be forced into the sample, and the spatial autocorrelation of PH and SOC, estimated by variogram fitting, can reduce the required sample size.

Workflow in brief:

Phase	Content
1	Set-up: parameters, frame and legend reading, grid and domain codes, correlation matrix
2	LUCAS points to be retained (forced, take-all stratum)
3	Variogram fitting on point survey data: national and per zone
4	SamplingStrata frame, precision constraints (CV = 5 %)
5	Initial solution by spatial K-means, domain by domain
6	Genetic algorithm (non-spatial baseline)
6b	Per-domain choice between non-spatial and spatial allocation
7	Sample selection, reports, figures, geopackages, strata polygons

2. Requirements and installation

Software. R (4.1 or later; RStudio recommended). The path examples below are Windows-style.

R packages loaded at the top of the script:

```
sp, raster, ggplot2, dplyr, tidyr, SamplingStrata, clhs,
sf, terra, readxl, gstat, parallel, doParallel
```

Automatic installation. A block at the top of the script (before the `library()` calls) installs what is missing, so no manual step is normally needed. It installs from CRAN the packages `sp, raster, ggplot2, dplyr, tidyr, clhs, sf, terra, readxl, gstat, doParallel` and `pak` if they are not yet available (`parallel` is part of base R).

SamplingStrata is handled separately: version 1.6 or later is required. If the package is missing or older, the script installs it from GitHub with `pak::pak("barcaroli/SamplingStrata")` (this needs an internet connection).

To install manually, run:

```
install.packages(c("pak","sp","raster","ggplot2","dplyr","tidyr","clhs",
                   "sf","terra","readxl","gstat","doParallel"))
pak::pak("barcaroli/SamplingStrata")
```

Hardware. The genetic algorithm runs in parallel (`parallel = TRUE`), using the available cores. The example frame has about 10,000 cells; run time depends on the machine and on the GA settings (Section 5.2).

In Phase 6b the script uses two internal, non-exported functions of SamplingStrata (`SamplingStrata:::buildStrataDFSpatial` and `:::bethel_cpp`). Another version of the package may behave differently or fail there; record the version you use.

3. Input data

All inputs are read from one folder (`Input/` in the repository). File names are hard-coded in the script; if you rename a file, edit the matching `read.csv / st_read / read_excel / rast` call.

File	Content	Required fields
frame_SML_1km_6_soil_units.csv	Sampling frame: one row per 1 km cell	POINT_ID, X_LAEA, Y_LAEA (EPSG:3035), ORIGINE, PH, PH_VAR, SOC, SOC_VAR, ARG, DEM, zona, soil_unit
Test_6_soil_units_Italy.gpkg	Legend: one polygon per soil unit	zona (zone name); CRS EPSG:3035
Soil_observed_points_2022_Italy.xlsx	LUCAS points, sheet *Points*	POINT_ID, X_WGS84, Y_WGS84, SOIL (1 = to be forced)
SharewithMS20260825-IT.csv	Point-level soil survey, used only to fit variograms	X_LAEA, Y_LAEA, pH_H2O, SOC
DEM_3035_1000m.tif	1 km elevation raster, EPSG:3035	Used for the variogram trend only (always read)

Notes on the frame:

- **ORIGINE**: cells with value `LUCAS_2km` are flagged as LUCAS-grid cells (`LUCAS_GRID = 1`).
- **POINT_ID** links LUCAS points to grid cells: LUCAS points must sit on grid nodes and share the ID.
- Domains are built as `zona*soil_unit`. The text before the asterisk is the key into the per-zone variogram tables, so zone names in the frame, in the legend `zona` field and in the survey join must match exactly.
- Cells with non-finite PH, SOC, ARG, DEM or variances are dropped automatically (a message reports how many).
- Auxiliary variables with negative values are shifted to be non-negative (a warning is issued).

4. Preparing the run

4.1 Set the folders

The three path variables near the top of the script **must be changed**: as delivered they point to a folder (`SML_sampling_procedure_example`) that differs from this repository.

```
inp_path <- "D:\\Google Drive\\ISPRA\\SML_sampling\\Input\\"
out_path <- "D:\\Google Drive\\ISPRA\\SML_sampling\\Output\\"
kriging_survey_path <- "D:/Google Drive/ISPRA/SML_sampling/Input/SharewithMS20260825-IT.csv"
```

Keep the trailing separator on `inp_path` and `out_path`, because file names are appended with `paste0()` (forward slashes also work). **The output folder must already exist**: the script does not create it. The repository tracks the folder as lower-case `output/`; on case-sensitive systems use exactly that spelling.

4.2 Run it

Open `SML_sampling.Rproj`, open the script and either press *Source* or run it section by section. From a console:

```
source("SML_sampling_example_with_6_Italy_soil_units.R", echo = TRUE)
```

The genetic algorithm uses `set.seed(1234)`, so the run is meant to be reproducible; change the seed to explore the variability of the solution. Progress is shown through console messages (LUCAS summary, correlation matrix, variogram fits, GA progress, Phase 6b results).

5. Customising the script

5.1 Main parameters

The block "Parameters & settings" (about line 150) groups the choices users are expected to change.

Parameter	Default	Meaning / effect
min_point_per_domain	5	Domains with fewer cells are dropped (too small to stratify or cluster).
use_spatial_fit	TRUE	TRUE: Phase 6b compares, per domain, the real kriging-based fitting/range with the non-spatial setting and keeps the spatial one only if it needs no more units. FALSE: non-spatial everywhere (fitting = 1, range = 0); no comparison. Use FALSE when local variograms are unreliable.
fitting_warn_threshold	0.9	A variogram R2 below this value triggers a warning in the summary report.
vars_cont_y	PH, SOC	Target variables (frame column names).
variance_cont_y	PH_VAR, SOC_VAR	Prediction variance of each target; same order as vars_cont_y.
vars_cont_x	ARG, DEM	Auxiliary variables used to build strata.
use_lucas_points	TRUE	TRUE: LUCAS points with SOIL == 1 are forced into the sample as a take-all stratum. FALSE: skip Phase 2.
use_lucas_grid	TRUE	TRUE: the LUCAS_GRID flag is carried in the frame, and sampled cells on the LUCAS grid are labelled *sampled_lucas*.
kriging_source_col	c(PH="pH_H2O", SOC="SOC")	Maps each target to the corresponding column of the point survey.
base_nome	Example_with_6_soil_units	Prefix for all output files.

5.2 Further settings inside the code

Setting (where)	Default	Meaning
min_obs_district (Phase 3)	30	Valid survey points a zone needs for its own variogram; otherwise the national model is used.
trend_covars (Phase 3)	DEM for PH and SOC	Trend covariates added to x + y in the regression-kriging variogram. Columns must exist in the survey (after the DEM extraction).
range_frac_max (fit_variogram_model)	0.8	Cap on the fitted range as a fraction of the largest empirical lag.
cv data frame (Phase 4)	0.05	Precision constraint (CV). Edit the `rep(0.05, ndom)` inside `cv_cols` to change it; the data frame has one column per target (CV1, CV2) and one row per domain, so values can differ by domain and by variable.
max_k (Phase 5)	5	Upper bound of K-means clusters (initial strata) per domain.
iter, pops (Phase 6)	15, 10	GA generations and population size. Increase for a more thorough search (longer run time).
parallel (Phase 6)	TRUE	Set FALSE if the parallel back-end causes problems.
kappa	1	Passed to SamplingStrata functions; leave unchanged unless you know the package's spatial method.
res_m (final export)	1000	Cell size in metres for the strata polygons; must match the grid.

5.3 Adapting the script to another area or other variables

1. Build a new frame CSV with the same field names (or change the names in the code: `X_LAEA`, `Y_LAEA`, `ORIGINE`, `zona`, `soil_unit`, `POINT_ID`).
2. Update the legend `.gpkg` so that its `zona` values match the frame.
3. Provide the survey with the columns named in `kriging_source_col` and the coordinates `X_LAEA`, `Y_LAEA`.
4. To use other targets or auxiliaries, change `vars_cont_y`, `variance_cont_y`, `vars_cont_x`, `kriging_source_col` and `trend_covars`. Also update `cor_vars` (correlation matrix) and the Phase 6b column list `c("ID","X1","LON","LAT","Y1","Y2","VAR1","VAR2","DOMAINVALUE")`, which is written for exactly two targets (Y1, Y2).
5. Update titles and text that mention PH, SOC, ARG, DEM, the two zones and "6 domains" in the report and figures, if needed.
6. Change `base_nome` so earlier results are not overwritten.

The LUCAS step requires the point `POINT_ID` to appear in the frame. Points that do not match (outside the study area) or have missing values in the target, variance or auxiliary variables are discarded and counted in the console message.

6. Outputs

Files are written to `out_path`, with prefix `base_nome`. Start with the summary report.

File	Content
_summary.txt	Text report: variogram fits (national and per zone), non-spatial optimisation, spatial comparison (Phase 6b), warnings, final sample, expected CVs, sites per domain.
_units.csv	One row per domain: number of cells, validity flag, total sites, LUCAS-grid sites, forced LUCAS sites.
_allpoints.csv	All selected points (sampled + forced LUCAS): cell, domain, coordinates (EPSG:3035), stratum, source.
_sample.csv	SamplingStrata sample with sampling weights.
_figures.pdf	CV boxplot, map by soil unit, stacked barplot of points per domain, geographic distribution of points.
_variograms.pdf	Empirical variograms and fitted models, national and per zone.
_correlation_matrix.pdf	Correlation matrix of PH, SOC, ARG, DEM.
.gpkg	The 1 km grid with domain and sample counts per domain (GeoPackage).
.RData	Complete workspace, for later inspection.
strata_SML.gpkg	Optimised strata as polygons (fields *soil_unit*, *STRATO*, *stratum_id*); written without the prefix.

In the *source* column of `_allpoints.csv`: **sampled** = newly selected cell; **sampled_lucas** = selected cell lying on the LUCAS grid; **lucas** = existing LUCAS point forced into the sample.

6.1 How to read the results

- **Sample size.** Section 2 of the summary gives the non-spatial sample per domain; section 3 compares it with the spatial one and shows which was kept. The final total in section 5 includes forced LUCAS points.
- **Expected CV.** The maximum over domains for PH and SOC should not exceed 0.05 (the boxplot in *_figures.pdf* has a red line at 0.05).
- **Warnings.** Section 4 flags low variogram fitting and use of the national model at zone level. In the reference run all four zone fits fell back to the national model (R2 0.41 for PH, 0.49 for SOC; ranges 109 and 80 km), because few survey points lie in the study area. The sample-size gain of the spatial version (about 28 %) depends on those national parameters and should be treated with caution; setting `use_spatial_fit <- FALSE` gives the conservative non-spatial allocation.
- **CV basis.** CVs are computed on the model-predicted values and variances in the frame, not on new field data.

6.2 Using the strata in the SML tool

Copy `strata_SML.gpkg` to the `Files/Strata/` folder of the SML tool and enter the settings in `parameters_values_for_SML_tool.txt`: soil unit file *Test_6_soil_units_Italy.gpkg* with ID field *soil_unit*; optional stratification from an existing polygon file (*strata_SML.gpkg*), column *stratum_id*, with areas not covered by the strata forming a separate stratum; point placement "simple random within strata".

7. Troubleshooting

Symptom	Likely cause and remedy
cannot open file / cannot find path	Paths not updated, missing trailing separator, or the output folder does not exist (Section 4.1).

Symptom	Likely cause and remedy
There is no package called ...	Install the package (Section 2).
stop(): fit_variogram_model() rejected the NATIONAL fit	The national variogram has R2 <= 0 for a variable. Inspect the survey data, the trend and the column in kriging_source_col; there is no further fallback.
Zone falls back to the national model	Fewer than min_obs_district valid points in the zone or a degenerate fit. Expected in the example; set use_spatial_fit to FALSE if the national model is not trusted locally.
optimStrata refuses to run / non-finite values	Missing or infinite values in X, Y, variance or coordinates; the script removes them, but check the frame if many are dropped.
"nStrata is not feasible"	A domain is too small; increase min_point_per_domain (the script already caps strata at half the cells of the domain).
Error in the Phase 6b ::: lines	Different SamplingStrata version; the internal functions changed. Use the version the script was written with.
Parallel errors / run too slow	Set parallel = FALSE, or reduce iter and pops for tests.
Slightly different totals between runs or machines	Independent Bethel calls differ by solver noise (about 1 % in the reference run); keep the seed fixed to reproduce.

Quick checklist: (1) paths updated, (2) output folder exists, (3) packages installed, (4) parameters set, (5) script run, (6) summary and warnings read, (7) figures checked, (8) strata copied to the SML tool.